Reg. No. | | | | | | | | | |

Question Paper Code : 15730

B.E. / B.TECH. DEGREE EXAMINATION, NOVEMBER / DECEMBER 2019

Third Semester

B.E. – Computer Science and Engineering

15CS301 – DATA STRUCTURES AND ALGORITHMS

(Regulations: Mepco - R2015)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

BTL, CO **PART A – (10 × 2 = 20 Marks)**

U, CO1 1. What is the recurrence relation to find out the number of multiplications and the initial condition for finding the n-th factorial of a number?

A, CO1 2. Show that $10n^2 + 4n + 2$ is $O(n^2)$.

A, CO2 3. The result evaluating the postfix expression 10 5 + 60 6 / * 8 – is _____. Justify.

A) 284 B) 142 C) 213 D) 71

A, CO2 4. What does the following function do for a given Linked List with first node as head?

```
void fun1(node* head)
{
 if(head = = NULL)
        return;
  fun1(head->next);
 cout<< head->data<<endl;
}
```

A, CO3 5. In a binary tree with n nodes, every node has an odd number of descendants. Every node is considered to be its own descendant. What is the number of nodes in the tree that have exactly one child? Justify.

A) 0 B) 1 C) n D) n-1

A, CO3 6. While inserting the elements 71, 65, 84, 69, 67, 83 in an empty Binary Search Tree (BST) in the sequence shown, the element in the lowest level is _____. Justify your answer.

A) 65 B) 67 C) 98 D) 81

R, CO4 7. List down the properties of an equivalence class.

R, CO4 8. Write down the advantages of Separate Chaining.

U, CO5 9. Consider the following specification of a graph G. Write its adjacency matrix.

V(G) = {1,2,3,4}

E(G) = {(1,2),(1,3),(3,3),(3,4)(4,1)} where V defines set of vertices and E defines set of edges.

U, CO5 10. What is the maximum number of edges in an acyclic undirected graph with n vertices? Justify.

A) N B) N-1 C) N+1 D) 2N-1

PART B – (5 × 16 = 80 Marks)

A, CO1 11. a) i. Design a recursive algorithm for computing the sum of first n cubes:

$S(n) = 1^3 + 2^3 + 3^3 + \ldots + n^3$. Analyze the algorithm mathematically. (8 Marks)

A, CO1 11. a) ii. Write a non-recursive algorithm to check whether all the elements in a given array of n elements are unique. Derive the time complexity of your algorithm. (8 Marks)

OR

A, CO1 11. b) i. Design a non-recursive algorithm to perform linear search in a given set of numbers and analyze its time complexity. (8 Marks)

A, CO1 11. b) ii. Solve the recurrence relation using Backward substitution method for the following:

A) X(n) = X(n/2) + 1, for n>1 and x(1) = 1

B) X(n) = X(n-1) + 1, for n>1 and x(1) = 0 (8 Marks)

A, CO2 12. a) i. Write an algorithm to reverse a given Doubly Linked List. (8 Marks)

A, CO2 12. a) ii. Enqueue the elements 10,15,20,5,9 into the circular queue of size 5 and then dequeue 2 elements out of the queue. In the resultant queue, insert 18 and 16.

Note: Show the detailed diagram for each step with front and rear index. (8 Marks)

OR

A, CO2 12. b) i. Implement a stack using linked list. Show both the PUSH and POP operations. (8 Marks)

A, CO2 12. b) ii. Write an algorithm to perform the following List ADT operations using arrays:

A) Insert at First

B) Delete at First

C) Insert at Last

D) Delete at Last (8 Marks)

A, CO3 13. a) i. Make a Binary Search Tree for the following sequence of numbers 45,32,90,34,68,72,15,24,30,66,11,50,10

Traverse the Binary Search Tree created in Preorder, Inorder and Postorder. (8 Marks)

U, CO3 13. a) ii. Illustrate the concept of Binary Min heap and its operations with example. (8 Marks)

OR

A, CO3 13. b) i. A complete binary tree is stored using an array. Write an algorithm to print the value of the binary tree in an ascending order.

(Example: Array value is 4,2,5,1,3 and output is 1,2,3,4,5) (8 Marks)

U, CO3 13. b) ii. Explain how the expression tree can be constructed with an example of your own. (8 Marks)

A, CO4 14. a) i. Show the result of the following sequence of instructions: Union (5, 6), Union (7, 8), Union (5, 7) when unions is performed arbitrarily and show the implicit representation of the above tree. (8 Marks)

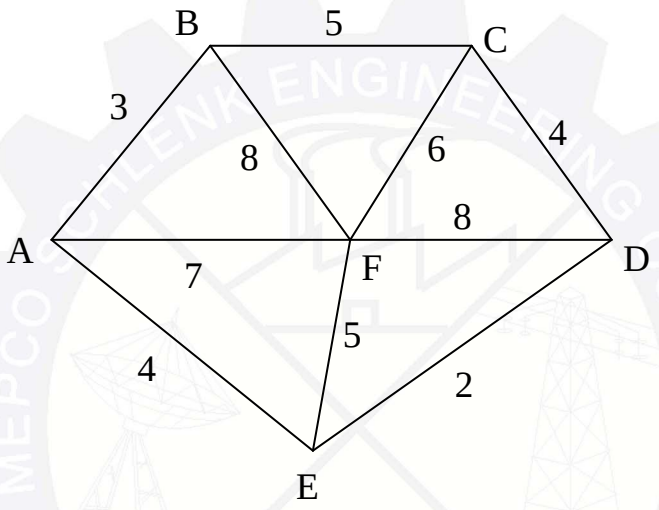
U, CO4 14. a) ii. Explain the concept of path comparison with an example. (8 Marks)

OR

A, CO4 14. b) i. Given input {4371, 1323, 6173, 4199, 4344, 9679, 1989} and a hash function h(x) = X(mod 10), show the result of separate chaining hash table and open addressing hash table using quadratic probing. (8 Marks)

U, CO4 14. b) ii. Discuss how hashing can be done using rehashing and extensible hashing techniques. (8 Marks)

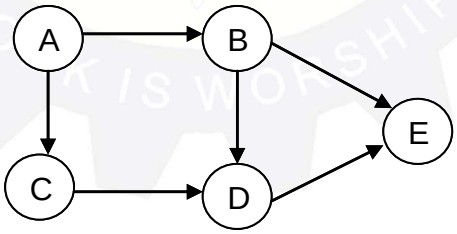
A, CO5 15. a) Write the Prim's algorithm and also find the minimum spanning tree using Prim's algorithm for the given graph.



(16 Marks)

OR

A, CO5 15. b) Design BFS and DFS algorithm for graph traversal and using it traverse the following directed graph. (Starting node : A)



(16 Marks)